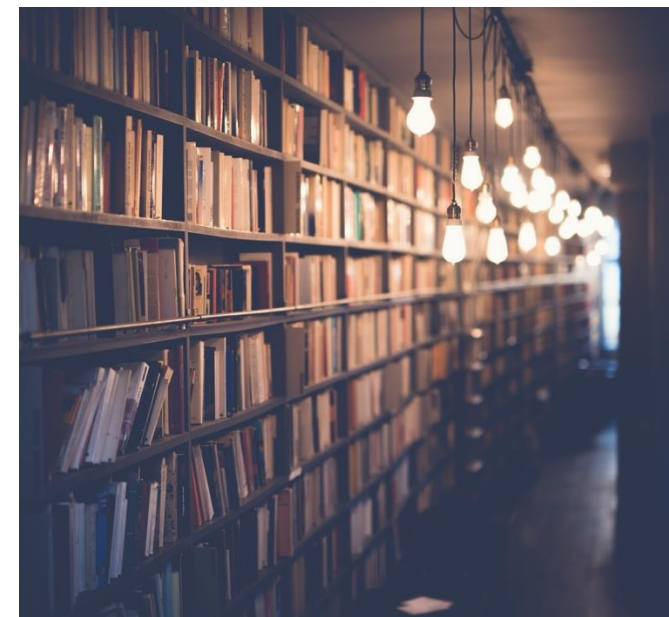




UiO : **CEMO** – Centre for Educational Measurement
University of Oslo

Meta-Analysis in Education

From Theory to Practice with R:
Multilevel Meta-Analysis



Ronny Scherer & Diego G. Campos

Workshop at the TU Dortmund
20./21.02.2025



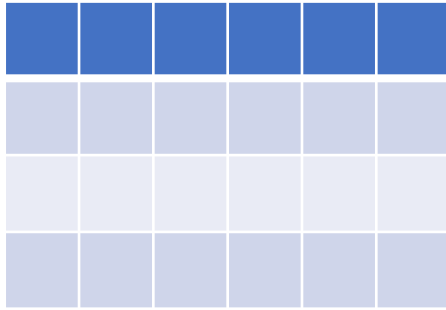
BEGIN.

Meta-Analytic Workflow

It is all about selecting appropriate working models

Selecting a Working Model

Meta-analytic data set



Nature of effect size multiplicity
and the assumptions behind it



Selection of a **Meta-Analytic
Working Model**

- Number of primary studies m
- Number of samples n
- Number of effect sizes k
- ...



Subsequent analyses

- Weighted average effect size
- Heterogeneity
- Moderator analyses
- Meta-regression
- Publication bias
- Outlier analyses
- ...

Possible Criteria for Model Selection

Proportion of Variance Components

A well-fitting model should have a reasonable distribution of variance across the levels. Small proportions of variance components suggest that this level may not be needed.

Information Criteria

Lower values of the AIC or BIC suggest a better-fitting model.

Likelihood-Ratio Test (LRT)

Direct comparison of nested models. An insignificant LRT suggests that the simpler model (with fewer levels) might be sufficient.

Conceptual Considerations

Nature of effect size multiplicity, research questions and goals.

Multilevel Meta-Analytic Models

Model heterogeneity at different levels in the presence of **hierarchical effect size multiplicity**

Three-Level Hierarchical Structure

(CMM, 2019)

Effect sizes nested in primary studies

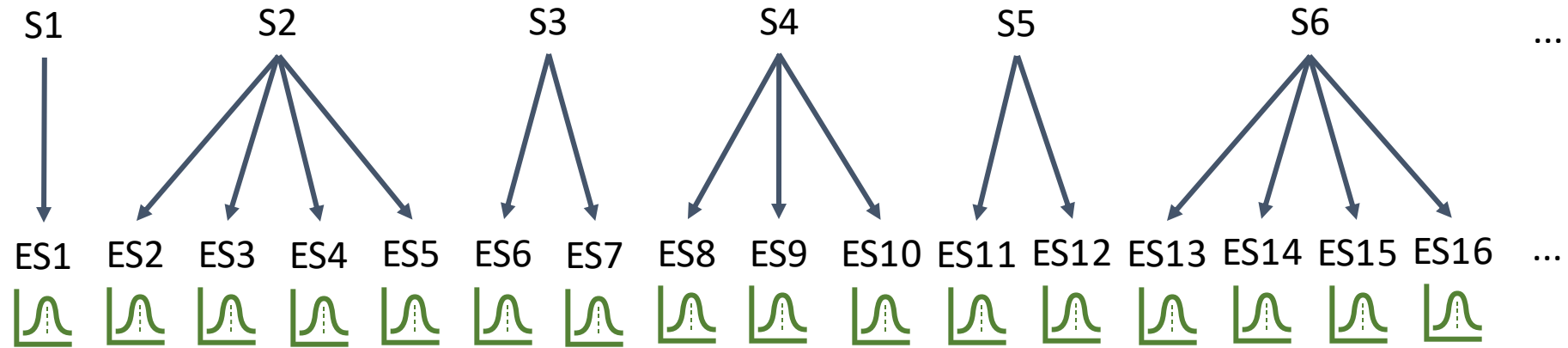
Studies (S)

Between-study variance $\tau_{(3)}^2$

Within-study variance $\tau_{(2)}^2$

Effect sizes (ES)

Sampling variance v_i



Typical meta-analytic structure if multiple independent samples are included in primary studies.

Multilevel Meta-Analysis (MLMA)

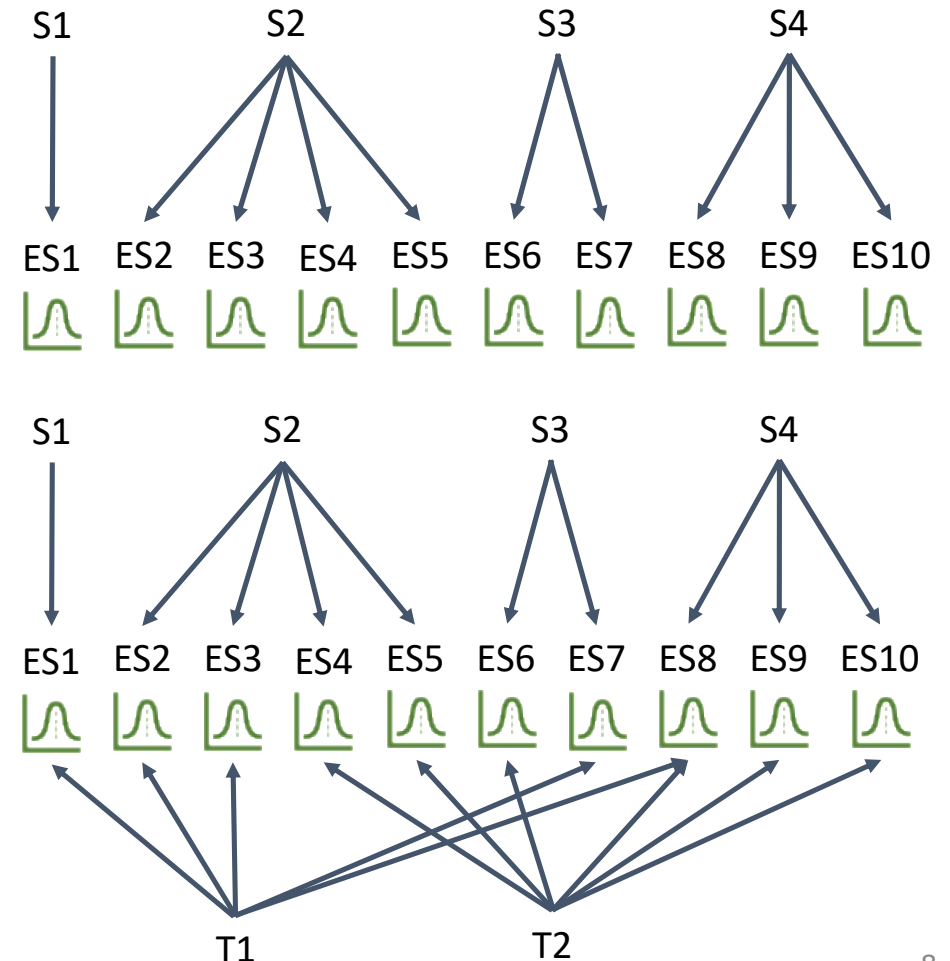
Model the hierarchical structure

- Effect sizes y_{ij} , $i = 1, \dots, k$; $j = 1, \dots, m$
- Sampling variances v_{ij}
- Multiple effect sizes nested in studies (usually from different samples)

Hierarchical and/or non-hierarchical data structures

Main idea

Model the different levels of analysis and the respective variance components





Data Example: Bilingualism and Cognition

Meta-analysis of the association between bilingualism and executive functioning (EF)

Lehtonen et al. (2018)

- Comparisons between adult bilinguals and monolinguals
- Hedges' g (bi-mono)
- 30 EF tasks
- 152 studies
- 891 effect sizes



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Psychological Bulletin

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Basque Center on Cognition, Brain and Language,
Donostia, Spain

Jan Antfolk
Abo Akademi University and University of Turku

Multilevel Meta-Analysis (MLMA)

(Cheung, 2015)

Three-level random-effects model (3LREM)

i : Effect sizes, j : Studies

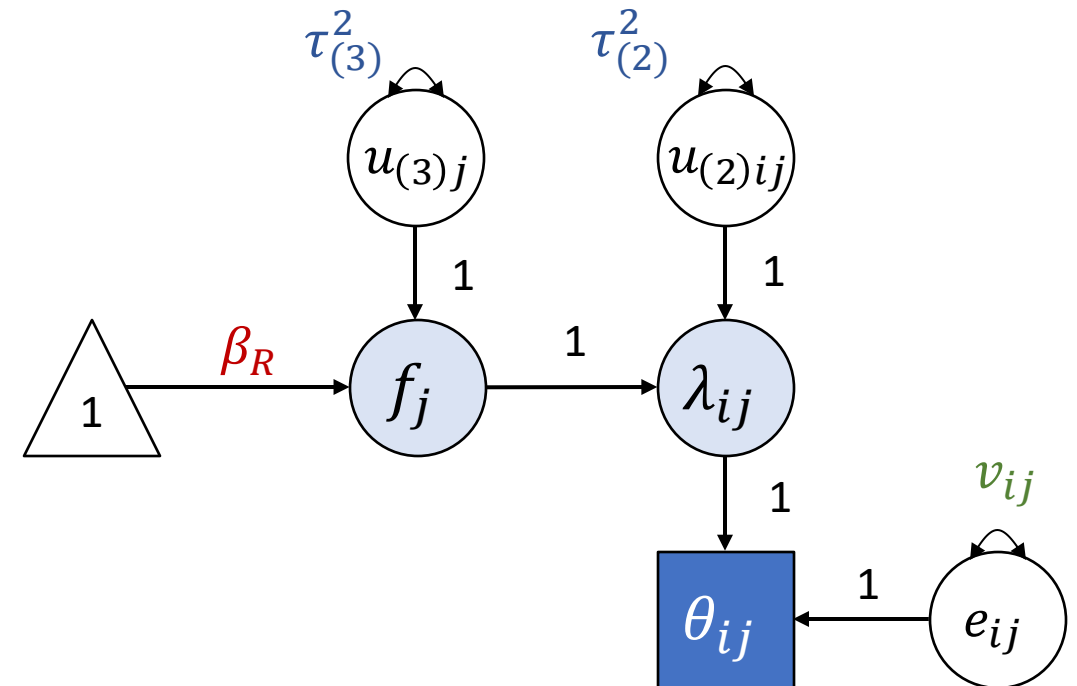
Level 1: $\theta_{ij} = \lambda_{ij} + e_{ij}$
 $e_{ij} \sim N(0, v_{ij})$

Level 2: $\lambda_{ij} = f_j + u_{(2)ij}$
 $u_{(2)ij} \sim N(0, \tau_{(2)}^2)$

Level 3: $f_j = \beta_R + u_{(3)j}$
 $u_{(3)j} \sim N(0, \tau_{(3)}^2)$

Total: $\theta_{ij} = \beta_R + u_{(2)ij} + u_{(3)j} + e_{ij}$

Path diagram:



β_R : Weighted average population effect size under the 3LREM

Multilevel Meta-Analysis (MLMA)

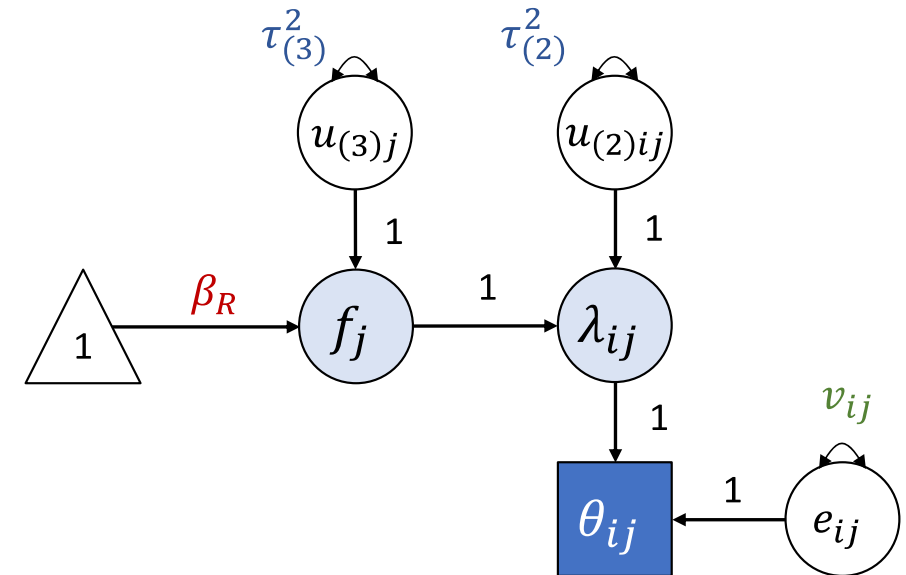
Univariate three-level **random-effects** model (3LREM)

Assumptions:

- Independence of random effects at different levels and sampling error
$$\text{Cov}(u_{(2)ij}, u_{(3)j}) = 0$$
$$\text{Cov}(u_{(2)ij}, e_{ij}) = 0$$
$$\text{Cov}(e_{ij}, u_{(3)j}) = 0$$
- Independence of L2 random effects within clusters and L3 between clusters

$$\text{Cov}(u_{(2)ij}, u_{(2)kj}) = 0$$
$$\text{Cov}(u_{(3)j}, u_{(3)m}) = 0$$

Path diagram:



β_R : Weighted average population effect size under the 3LREM

Multilevel Meta-Analysis (MLMA)

Univariate three-level **random-effects** model (3LREM)

Procedures:

- **Hypothesis testing** via model comparisons (e.g., information criteria, likelihood-ratio tests)

$$\tau_{(2)}^2 = \tau_{(3)}^2, \tau_{(2)}^2 = 0, \tau_{(3)}^2 = 0$$

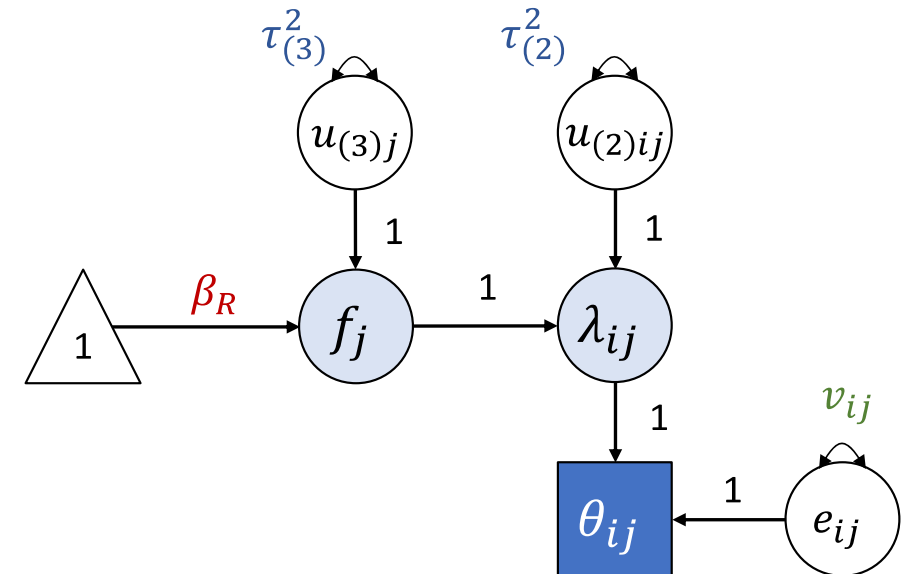
- Level-specific **heterogeneity indices**:

$$I_{(2)}^2, I_{(3)}^2$$

- Level-specific **variance explanations**:

$$R_{(2)}^2, R_{(3)}^2$$

Path diagram:



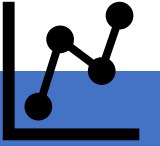
β_R : Weighted average population effect size under the 3LREM



Data Example: Bilingualism and Cognition

Three-level hierarchical structure: Identifying variables (IDs) are the key.

StudyName	Sample	Domain	ESID	StudyID	SampleID	TaskID	g	vg
Lee & Chan (2000)	1	Inhibition	1	78	780001	20	-0,26	0,05
Rosselli et al. (2000)	1	Fluency	2	116	1160001	3	-0,7	0,12
Rosselli et al. (2000)	1	Fluency	3	116	1160001	10	0,02	0,11
Gollan et al. (2002)	1	Fluency	4	54	540001	3	-0,87	0,07
Gollan et al. (2002)	1	Fluency	5	54	540001	10	-0,39	0,07
Rosselli et al. (2002)	1	Inhibition	6	117	1170001	20	-0,5	0,11
Rosselli et al. (2002)	1	Monitoring	7	117	1170001	20	-0,12	0,11
Boutla (2003)	1	WM	8	27	270001	17	0,78	0,14
Bialystok et al. (2004)	1	Inhibition	9	9	90001	14	2,87	0,45
Bialystok et al. (2004)	1	Monitoring	10	9	90001	14	1,3	0,25
Bialystok et al. (2004)	2	Inhibition	11	9	90002	14	1,03	0,23
Bialystok et al. (2004)	2	Monitoring	12	9	90002	14	1,06	0,23
Bialystok et al. (2004)	3	Inhibition	13	9	90003	14	1,73	0,09
Bialystok et al. (2004)	3	Monitoring	14	9	90003	14	4,63	0,24
Bialystok et al. (2004)	3	WM	15	9	90003	11	-0,17	0,06
Bialystok et al. (2004)	3	WM	16	9	90003	11	-0,23	0,06



Data Example: Bilingualism and Cognition

Three-level REM

```
# Model estimation
REM3st <- rma.mv(g,
  vg,
  data = lehtonen18,
  method = "REML",
  tdist = TRUE,
  test = "t",
  random = list(~ 1 | StudyID/ESID))

# Summary of the results
summary(REM3st)
```

Level 1: Sampling variation

Level 2: Within-study between-effects variation

Level 3: Between-study variation

```
## Multivariate Meta-Analysis Model (k = 891; method: REML)
##
##      logLik      Deviance      AIC      BIC      AICc
## -696.1839  1392.3678  1398.3678  1412.7415  1398.3949
##
## Variance Components:
##
##      estim      sqrt      nlvls      fixed      factor
## sigma^2.1  0.1237  0.3518      152      no      StudyID  $\tau_{(3)}^2$ 
## sigma^2.2  0.1159  0.3405      891      no      StudyID/ESID  $\tau_{(2)}^2$ 
##
## Test for Heterogeneity:
## Q(df = 890) = 3239.5972, p-val < .0001
##
## Model Results:
##
## estimate      se      tval      df      pval      ci.lb      ci.ub      *
## 0.0766  0.0343  2.2324  890  0.0258  0.0093  0.1439
```



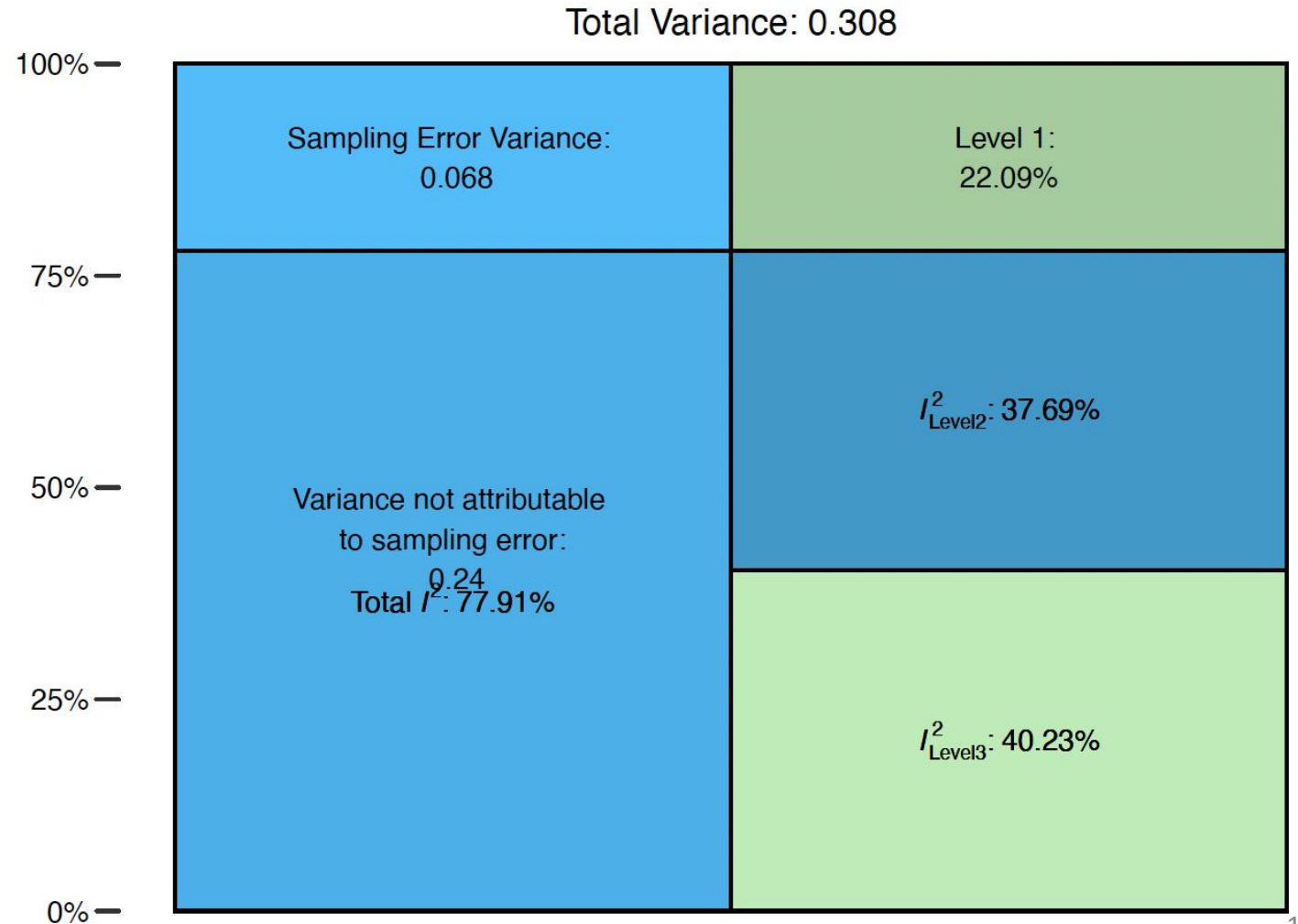
Data Example: Bilingualism and Cognition

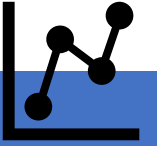
Three-level REM

```
# Total variance distribution  
dmetar::mlm.variance.distribution(x = REM3st)
```

```
##           % of total variance    I2  
## Level 1           22.08602    ---  
## Level 2           37.68807  37.69  
## Level 3           40.22591  40.23  
## Total I2: 77.91%
```

```
# Extract the variance components  
res <- dmetar::var.comp(REM3st)  
plot(res)
```





Data Example: Bilingualism and Cognition

Three-level REM with cluster-robust standard errors

```
# Cluster-robust standard errors
coef_test(REM3st, vcov = "CR2")
```

```
metafor::robust.rma.mv(REM3st,
  cluster = StudyID,
  clubSandwich = TRUE,
  adjust = TRUE)
```

```
##      Coef. Estimate      SE t-stat d.f. (Satt) p-val (Satt) Sig.
## intrcpt    0.0766 0.0343    2.23      142    0.0271    *
```

```
## Model Results:
##
## estimate      se1    tval1      df1    pval1    ci.lb1    ci.ub1
##    0.0766 0.0343    2.2325    142.32    0.0271    0.0088    0.1444    *
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## 1) results based on cluster-robust inference (var-cov estimator: CR2,
##    approx t-test and confidence interval, df: Satterthwaite approx)
```



Data Example: Bilingualism and Cognition

Overview of pooled effect sizes and variance estimates

Results	Ignore. Standard REM	Aggregate. Borenstein method	Correct. RVE Correlated	Correct. RVE Hierarchical	Model. 3LREM
$\bar{g}$	0.076	0.060	0.058	0.074	0.077
95 % CI	[0.041, 0.111]	[-0.011, 0.131]	[-0.149, 0.131]	[0.101, 0.138]	[0.009, 0.144]
Variances	$\tau^2=0.207$ (between effect sizes)	$\tau^2=0.143$ (between studies)	$\tau^2=0.264$ (between studies)	$\tau^2=0.173$ (between studies) $\omega^2=0.009$ (within studies)	$\tau^2_{(3)}=0.124$ (between studies) $\tau^2_{(2)}=0.116$ (within studies)



Data Example: Bilingualism and Cognition

Meta-analysis of the association between bilingualism and executive functioning (EF)

Lehtonen et al. (2018)

- Comparisons between adult bilinguals and monolinguals
- Hedges' g (bi-mono)
- 30 EF tasks
- 152 studies
- 891 effect sizes



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Basque Center on Cognition, Brain and Language,
Donostia, Spain

Jan Antfolk
Abo Akademi University and University of Turku

Activity: A Bit of Reflection

Reflect

Consider the information about the Lehtonen et al. (2018) meta-analysis again.

Which other structure(s) might describe the meta-analytic data?





Data Example: Bilingualism and Cognition

Four-level REM with a hierarchical structure

```
# Model estimation
REM3sast <- rma.mv(g,
  vg,
  data = lehtonen18,
  method = "REML",
  tdist = TRUE,
  test = "t",
  random = list(~ 1 | StudyID/SampleID/ESID))

# Summary of the results
summary(REM3sast)
```

```
## Multivariate Meta-Analysis Model (k = 891; method: REML)
```

```
##
```

	logLik	Deviance	AIC	BIC	AICc
	-693.8499	1387.6998	1395.6998	1414.8647	1395.7450

```
##
```

```
## Variance Components:
```

```
##
```

	estim	sqrt	nlvls	fixed	factor	τ^2
## sigma^2.1	0.1116	0.3341	152	no	StudyID	$\tau^2_{(3)}$
## sigma^2.2	0.0178	0.1335	273	no	StudyID/SampleID	$\tau^2_{(2)}$
## sigma^2.3	0.1078	0.3283	891	no	StudyID/SampleID/ESID	$\tau^2_{(4)}$

```
##
```

```
## Test for Heterogeneity:
```

```
## Q(df = 890) = 3239.5972, p-val < .0001
```

```
##
```

```
## Model Results:
```

```
##
```

	estimate	se	tval	df	pval	ci.lb	ci.ub	
##	0.0778	0.0340	2.2870	890	0.0224	0.0110	0.1446	*

β_R

Cross-Classified Data Structure

(CMM, 2019)

Four-level non-hierarchical structure with cross-classification

Studies (S)

Between-study variance $\tau_{(3)}^2$

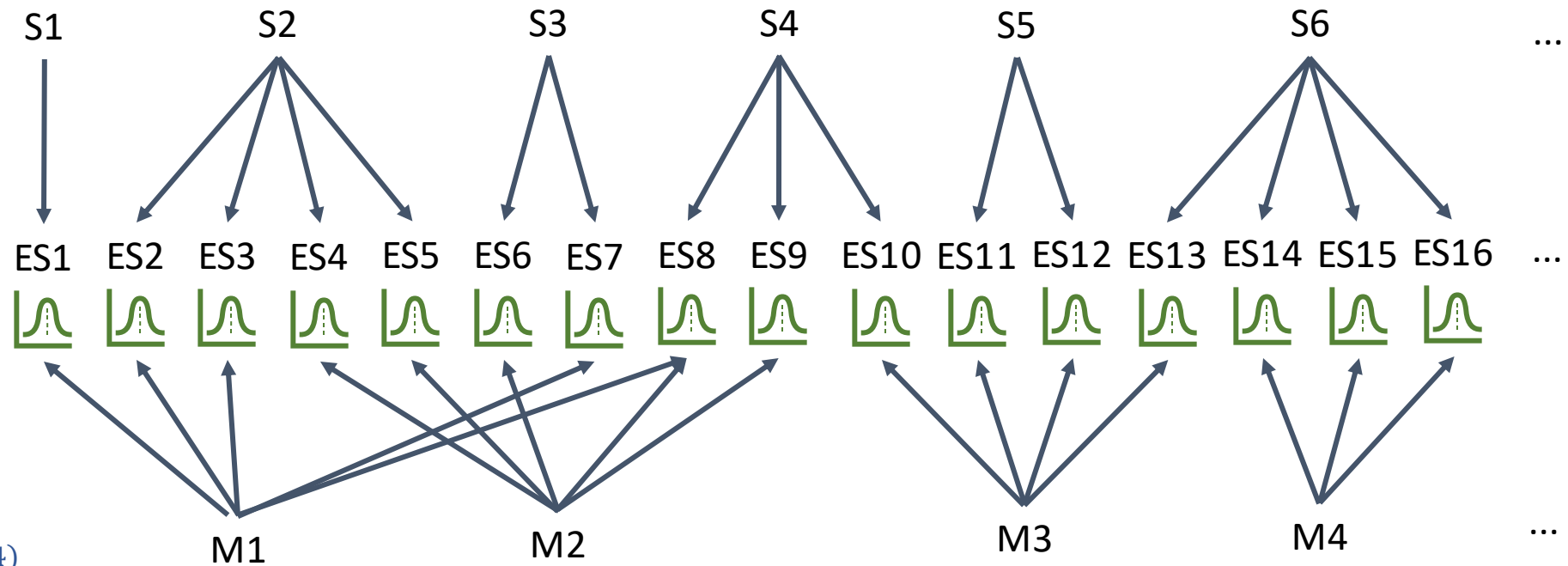
Within-study variance $\tau_{(2)}^2$

Effect sizes (ES)

Sampling variance v_i

Measures (M)

Between-measure variance $\tau_{(4)}^2$



Typical meta-analytic structure if multiple measures are included.



Data Example: Bilingualism and Cognition

Four-level CCREM

```
# Model estimation
CCREM <- rma.mv(g,
  vg,
  data = lehtonen18,
  method = "REML",
  tdist = TRUE,
  test = "t",
  random = list(~ 1 | StudyID/ESID,
    ~ 1 | TaskID))

# Summary of the results
summary(CCREM)
```

Two independent levels of analysis:
Study level and task level

(For more details about CCREMs in meta-analysis, please refer to
Fernández-Castilla et al., 2019)

```
## Multivariate Meta-Analysis Model (k = 891; method: REML)
##
##      logLik      Deviance      AIC      BIC      AICc
## -680.0790    1360.1580    1368.1580    1387.3229    1368.2032
##
## Variance Components:
##
##              estim      sqrt  nlvls  fixed      factor
## sigma^2.1    0.1315    0.3627    152      no      StudyID
## sigma^2.2    0.1007    0.3173    891      no    StudyID/ESID
## sigma^2.3    0.0251    0.1584     30      no      TaskID
##
## Test for Heterogeneity:
## Q(df = 890) = 3239.5972, p-val < .0001
##
## Model Results:
##
## estimate      se      tval    df      pval      ci.lb      ci.ub
##    0.0598    0.0513    1.1648   890    0.2444    -0.0410    0.1606
```

$\tau_{(3)}^2$
 $\tau_{(2)}^2$
 $\tau_{(4)}^2$

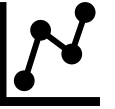
β_R



Data Example: Bilingualism and Cognition

Overview of pooled effect sizes and variance estimates

Results	Ignore. Standard REM	Aggregate. Borenstein method	Correct. RVE Correlated	Correct. RVE Hierarchical	Model. 3LREM	Model. CCREM
$\bar{g}$	0.076	0.060	0.058	0.074	0.077	0.060
95 % CI	[0.041, 0.111]	[-0.011, 0.131]	[-0.149, 0.131]	[0.101, 0.138]	[0.009, 0.144]	[-0.041, 0.160]
Variances	$\tau^2=0.207$ (between effect sizes)	$\tau^2=0.143$ (between studies)	$\tau^2=0.264$ (between studies)	$\tau^2=0.173$ (between studies) $\omega^2=0.009$ (within studies)	$\tau^2_{(3)}=0.124$ (between studies) $\tau^2_{(2)}=0.116$ (within studies)	$\tau^2_{(3)}=0.132$ (between studies) $\tau^2_{(2)}=0.101$ (within studies) $\tau^2_{(4)}=0.025$ (between tasks)

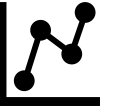


Information Criteria

AIC and BIC for the random-effects models

	Standard REM	Three-level REM	Four-level REM	CCREM
logLik	-780.17	-696.18	-693.85	-680.08
AIC	1564.35	1398.37	1395.70	1368.16
BIC	1573.93	1412.74	1414.86	1387.32

Result: Model CCREM shows the lowest AIC and BIC.



Likelihood-Ratio Tests of Nested Models

LRTs between the
random-effects models

(a) REM3st consistently
preferred over REM2

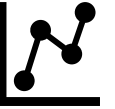
```
## Standard REM vs. three-level REM
anova(REM2, REM3st)
```

##		df	AIC	BIC	AICc	logLik	LRT	pval
##	Full	3	1398.3678	1412.7415	1398.3949	-696.1839		
##	Reduced	2	1564.3472	1573.9297	1564.3608	-780.1736	167.9794	<.0001

(b) REM3sast not consistently
preferred over REM3st

```
## Three-level REM vs. Four-level REM
anova(REM3st, REM3sast)
```

##		df	AIC	BIC	AICc	logLik	LRT	pval
##	Full	4	1395.6998	1414.8647	1395.7450	-693.8499		
##	Reduced	3	1398.3678	1412.7415	1398.3949	-696.1839	4.6680	0.0307



Likelihood-Ratio Tests of Nested Models

LRTs between the
random-effects models

(c) CCREM consistently preferred
over REM3st

```
## Three-level REM vs. cross-classified REM  
anova(REM3st, CCREM)
```

##		df	AIC	BIC	AICc	logLik	LRT	pval
##	Full	4	1368.1580	1387.3229	1368.2032	-680.0790		
##	Reduced	3	1398.3678	1412.7415	1398.3949	-696.1839	32.2098	<.0001

Result: Model CCREM may serve as a baseline model.

Model Selection

The **CCREM** seems a good choice for a baseline model if we assume

- a hierarchical nesting of effect sizes within studies
- a hierarchical nesting of effect sizes within tasks, and
- the independence of the study and task levels.

Notice that the CCREM quantifies variation between tasks and assumes that the **tasks are independent** of each other.

We will later see a model in which we assume that tasks as dependent (in the so-called «**correlated and hierarchical effects model**»).

Moderator Analyses

Mixed-effects meta-regression models to explain heterogeneity

Moderator Analysis

(Cheung, 2015)

Three-level mixed-effects meta-regression

i : Effect sizes, j : Studies

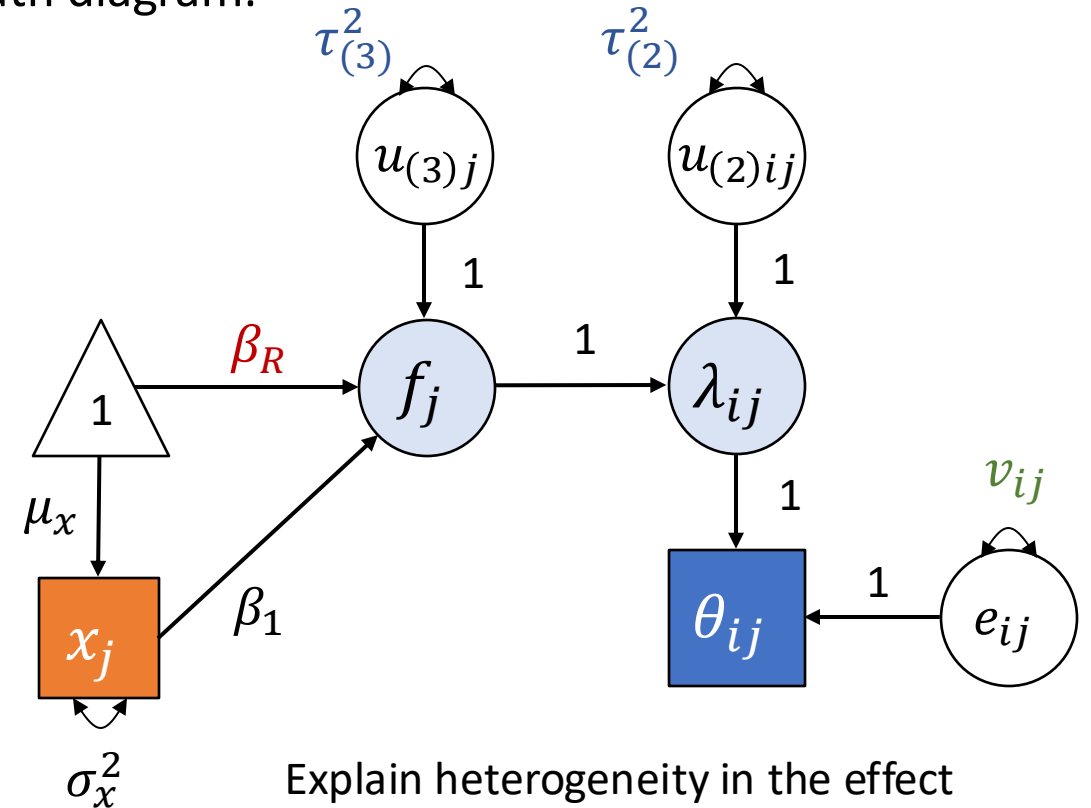
Level 1: $\theta_{ij} = \lambda_{ij} + e_{ij}$
 $e_{ij} \sim N(0, v_{ij})$

Level 2: $\lambda_{ij} = f_j + u_{(2)ij}$
 $u_{(2)ij} \sim N(0, \tau_{(2)}^2)$

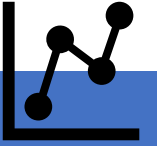
Level 3: $f_j = \beta_R + \beta_1 x_j + u_{(3)j}$
 $u_{(3)j} \sim N(0, \tau_{(3)}^2)$

Total: $\theta_{ij} = \beta_R + \beta_1 x_j + u_{(2)ij} + u_{(3)j} + e_{ij}$

Path diagram:



Explain heterogeneity in the effect sizes by a **moderating variable** x_j



Data Example: Bilingualism and Cognition

Four-level CCMEM

Model specification with moderators

```
# Model estimation
CCMEM <- rma.mv(g,
               vg,
               data = lehtonen18,
               method = "REML",
               tdist = TRUE,
               test = "t",
               random = list(~ 1 | StudyID/ESID,
                             ~ 1 | TaskID),
               mods = ~ Stimulus.binary + StudyType.binary)

# Summary of the results
summary(CCMEM)
```



Data Example: Bilingualism and Cognition

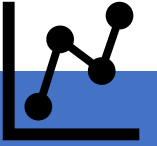
Four-level CCMEM

Model specification with moderators

```
# Model estimation
CCMEM <- rma.mv(g,
  vg,
  data = lehtonen18,
  method = "REML",
  tdist = TRUE,
  test = "t",
  random = list(~ 1 | StudyID/ESID,
    ~ 1 | TaskID),
  mods = ~ Stimulus.binary + StudyType.binary)

# Summary of the results
summary(CCMEM)
```

```
## Multivariate Meta-Analysis Model (k = 891; method: REML)
##
##      logLik    Deviance      AIC      BIC      AICc
## -671.0600  1342.1201  1354.1201  1382.8539  1354.2154
##
## Variance Components:
##
##              estim    sqrt  nlvls  fixed      factor
## sigma^2.1    0.1284  0.3583   152    no      StudyID
## sigma^2.2    0.0981  0.3132   891    no  StudyID/ESID
## sigma^2.3    0.0215  0.1467    30    no      TaskID
##
## Test for Residual Heterogeneity:
## QE(df = 888) = 3150.1797, p-val < .0001
##
## Test of Moderators (coefficients 2:3):
## F(df1 = 2, df2 = 888) = 7.6335, p-val = 0.0005
##
## Model Results:
##
##              estimate      se    tval   df    pval    ci.lb    ci.ub
## intrcpt           0.1139  0.0756   1.5077  888   0.1320  -0.0344   0.2622
## Stimulus.binary  -0.1725  0.0454  -3.7994  888   0.0002  -0.2616  -0.0834 ***
## StudyType.binary   0.0652  0.0742   0.8787  888   0.3798  -0.0804   0.2109
```



Data Example: Bilingualism and Cognition

Four-level CCMEM

Variance explanation (R^2)

Variance explanation as the **reduction of variance components** when comparing the model without moderators to the model with moderators

Key advantage of MLMA:

Find out where the variance is (explained).

```
# Between-study variance explanation  
max((CCREM$sigma2[1]-CCMEM$sigma2[1])/CCREM$sigma2[1], 0)
```

```
## [1] 0.02370672
```

```
# Within-study between-effects variance explanation  
max((CCREM$sigma2[2]-CCMEM$sigma2[2])/CCREM$sigma2[2], 0)
```

```
## [1] 0.02530506
```

```
# Between-task variance explanation  
max((CCREM$sigma2[3]-CCMEM$sigma2[3])/CCREM$sigma2[3], 0)
```

```
## [1] 0.1430961
```

Practice: Multilevel Meta-Analysis

Effectiveness of Computer Programming Interventions



Transfer Effects of Computer Programming

Meta-analytic data set by Scherer et al. (2019)

Data set

- 530 effect sizes
- 102 studies
- Pretest-posttest and posttest-only experimental-control group designs
- Hedges' g (EXP-CON)



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0022-0663/19/\$12.00

Journal of Educational Psychology

2019, Vol. 111, No. 5, 764–792
<http://dx.doi.org/10.1037/edu0000314>

The Cognitive Benefits of Learning Computer Programming: A Meta-Analysis of Transfer Effects

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Humboldt-Universität zu Berlin

To what extent are computer programming interventions effective for teaching cognitive skills?



Transfer Effects of Computer Programming

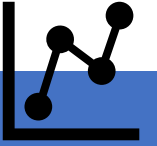
Meta-analytic data set by Scherer et al. (2019)

Data set

- 530 effect sizes
- 102 studies
- Pretest-posttest and posttest-only experimental-control group designs
- Hedges' g (EXP-CON)

Your tasks

1. Estimate the **overall effect size**,
 - (a) Ignoring the hierarchical data structure.
 - (b) Correcting for the hierarchical data structure.
 - (c) Modelling the hierarchical data structure.
2. Using a **three-level mixed-effects model**, examine whether `Design.binary` and `ControlGroup.binary` moderate the effects.



Transfer Effects of Computer Programming

Meta-analytic data set by Scherer et al. (2019)

To get started...

```
# Load packages
library(metafor)
library(robumeta)
library(clubSandwich)
```

```
# Read in the data and define them as an object
## csv-file
scherer19 <- read.csv2("Scherer-et-al-2019.csv",
                      header = TRUE,
                      dec = ",")
head(scherer19, 10)
```

```
# Transform some variables for later moderator analysis
## Study design
## Posttest-only CG coded as 0 and Pretest-Posttest CG as 1
scherer19$Design.binary <- as.numeric(factor(scherer19$Design))-1

## Control group
## treated coded as 0 and untreated coded as 1
scherer19$ControlGroup.binary <- as.numeric(factor(scherer19$ControlGroup))-1
```

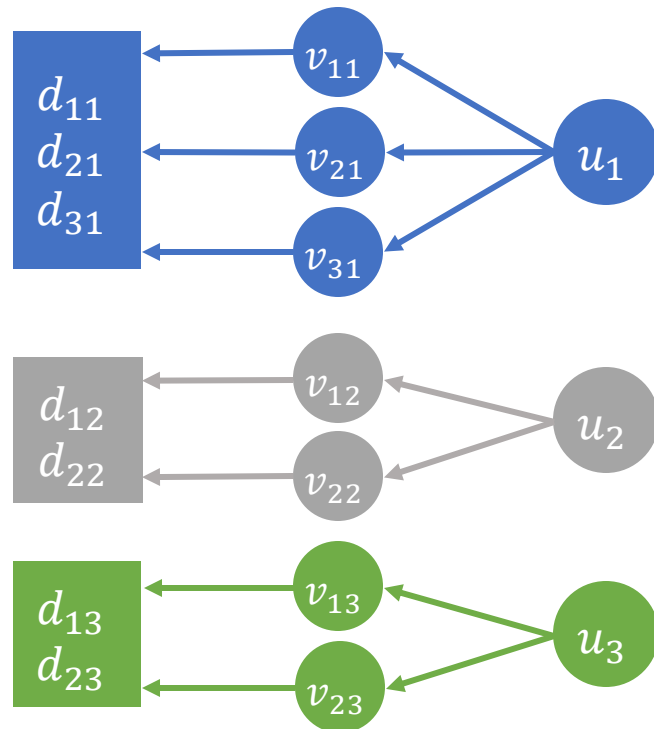
Multilevel Meta-Analysis with Correlated Effects

Probably, the most realistic situation in meta-analyses (correlated and hierarchical effects)

Models for Correlated and Hierarchical Effects

(Pustejovsky & Tipton, 2021, <https://doi.org/10.1007/s11121-021-01246-3>;
Illustration inspired by J. E. Pustejovsky, UseR! Oslo Talk, 02.09.2021)

Correlated *and* Hierarchical Effects (CHE)



Correlated effect sizes
Between-study heterogeneity
Within-study heterogeneity

Steps

1. Estimate the sampling covariance matrix
2. Specify and estimate the multilevel MA with information about the correlation (ρ)
3. Obtain robust standard errors

Strategy

Estimate variances at different levels &
Specify the correlation between effect sizes

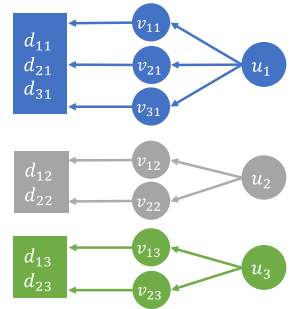
metafor
clubSandwich



Multilevel Meta-Analysis with CE (MLMA-CE)

Model correlated and hierarchical effects

- Effect sizes y_{ij} , $i = 1, \dots, k$; $j = 1, \dots, m$
- Sampling variances v_{ij} with average $v_{.j}$
- Constant sampling correlation ρ
- Within-study variation ω^2
- Between-study variation τ^2



$$y_{ij} = \beta_R + u_{(2)ij} + u_{(3)j} + e_{ij}$$

Main idea

Incorporate a constant sampling correlation in the model and estimate variances at multiple levels

Note: This model can be extended to a model in which the study-specific sampling covariance matrices are incorporated.

$$\text{Var}(u_{(3)j}) = \tau^2$$

$$\text{Var}(u_{(2)ij}) = \omega^2$$

$$\text{Var}(e_{ij}) = v_{.j}$$

$$\text{Cov}(e_{hj}, e_{ij}) = \rho v_{.j}$$



Data Example: Bilingualism and Cognition

MLMA-CE working model



Lehtonen et al. (2018)

- Comparisons between adult bilinguals and monolinguals
- Hedges' g (bi-mono)
- 30 EF tasks
- 152 studies
- 891 effect sizes

```
## We provide a constant sampling correlation (csc).  
rho <- 0.5
```

```
## Define the sampling variance-covariance matrix per study  
V_mat <- metafor::vcalc(lehtonen18$vg,  
                        cluster = lehtonen18$StudyID,  
                        rho = rho,  
                        checkpd = TRUE,  
                        nearpd = TRUE)
```

```
## Estimate the full model of the meta-analysis with three levels  
REM3L.csc <- rma.mv(g,  
                   V = V_mat,  
                   random = list(~ 1 | StudyID/ESID),  
                   data = lehtonen18,  
                   method = "REML",  
                   tdist = TRUE,  
                   test = "t")
```

```
## Summarize the results  
summary(REM3L.csc, digits=4)
```



Data Example: Bilingualism and Cognition

MLMA-CE working model

Model specification and robust standard errors

```
## RVE standard errors
conf_int(REM3L.csc, vcov = "CR2")
```

```
##      Coef. Estimate      SE d.f. Lower 95% CI Upper 95% CI
## intrcpt    0.0253 0.0329   131     -0.0398    0.0904
```

```
## Multivariate Meta-Analysis Model (k = 891; method: REML)
##
##      logLik    Deviance      AIC      BIC      AICc
## -701.9632  1403.9263  1409.9263  1424.3000  1409.9534
##
## Variance Components:
##
##              estim    sqrt  nlvls  fixed      factor
## sigma^2.1    0.0315  0.1775   152    no      StudyID
## sigma^2.2    0.2179  0.4668   891    no  StudyID/ESID
##
## Test for Heterogeneity:
## Q(df = 890) = 27400504814.6733, p-val < .0001
##
## Model Results:
##
## estimate      se    tval   df    pval    ci.lb    ci.ub
##    0.0253  0.0329  0.7687  890  0.4423  -0.0393  0.0899
```



Data Example: Bilingualism and Cognition

MLMA-CE working model
with moderator effects

Model specification and
robust standard errors

```
## Variance Components:
##
##               estim      sqrt  nlvls  fixed      factor
## sigma^2.1  0.0824  0.2870   152    no      StudyID
## sigma^2.2  0.1801  0.4244   891    no  StudyID/ESID
##
## Test for Residual Heterogeneity:
## QE(df = 888) = 4112.3027, p-val < .0001
##
## Test of Moderators (coefficients 2:3):
## F(df1 = 2, df2 = 888) = 11.4250, p-val < .0001
##
## Model Results:
##
##               estimate      se      tval    df      pval      ci.lb      ci.ub
## intrcpt           0.1093  0.0635   1.7198  888   0.0858   -0.0154   0.2340   .
## Stimulus.binary   -0.1847  0.0396  -4.6679  888  <.0001   -0.2624  -0.1071  ***
## StudyType.binary    0.0834  0.0735   1.1354  888   0.2565   -0.0608   0.2276

##               Coef. Estimate      SE d.f. Lower 95% CI Upper 95% CI
##               intrcpt    0.1093 0.0718 46.0      -0.0353      0.2539
## Stimulus.binary  -0.1847 0.0567 62.0      -0.2981     -0.0714
## StudyType.binary   0.0834 0.0794 82.9      -0.0745      0.2413
```

```
## RVE standard errors
conf_int(MEM3L.csc, vcov = "CR2")
```

Activity: Be the Expert.

Help needed!

Which sources of information about the correlation between multiple effect sizes would you recommend to your colleagues?

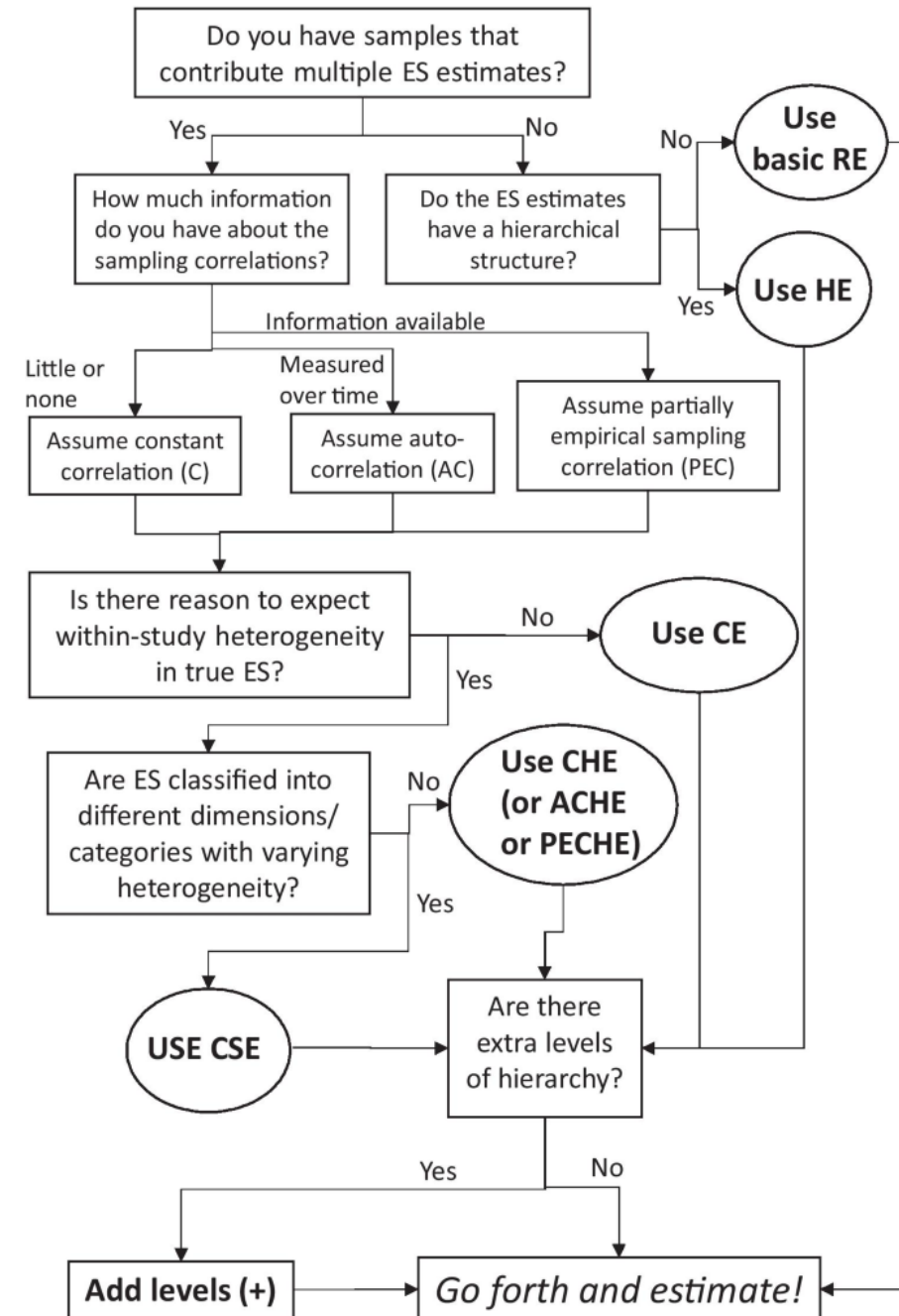


Decision Schemes

What is the nature of the multiple effects?

- Hierarchical effects
- Correlated effects
- Hierarchical and correlated effects

(Pustejovsky et al., 2021, <https://doi.org/10.1007/s1121-021-01246-3>)



The End.



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